COURSE

AQA GCSE Business (8132)

Break-even (financial calculations)

3.6 Finance · 3.6.3 Financial terms and calculations: break-even

Revision for 3.6 Finance: each firm works it out on its own deals.

Class	Topic
AQA GCSE Business (8132)	**Break-even (financial calculations)**

THE LESSON

Setup — 2 min

Round 1 — Decisions 3 min · Deals 5 min · Topic tasks 4 min

Firm decisions	Deals	TASK 1 Profit on the chart	TASK 2 Lower break-even

Round 2 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	Deals	TASK 3 Margin of safety	TASK 4 Profit on the deal	B2B deal

Round 3 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	Deals	TASK 5 Less risky deal	TASK 6 Profit at the sales	B2B deal

Final questions — 3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

Your break-even chart.

Profit at that output? Minus allowed.

WORK OUT THE NUMBER · EXAMPLE FIGURES

Cars sold	**10 cars**
Total revenue at that output	**£220,000**
Fixed costs	**£4,250**
Variable costs at that output	**£210,000**

METHOD

Profit = total revenue – fixed costs – variable costs

£220,000 – £4,250 – £210,000 = £5,750

Why: Profit is the gap between the revenue line and the total cost line.

Task 2

ROUND 1

You want to lower your break-even point.

Which two changes would lower it?

PICK TWO

- ✔ **Cut the fixed costs**
- ○ Pay higher rent
- ○ Pay more for each car
- ✔ **Raise the selling price**
- ○ Sell more cars

Why: More added per car or lower fixed costs; selling more does not move it.

Task 3

ROUND 2

Three charts: what you sell, and the break-even output.

Order by margin of safety, largest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **Sell 14 cars, break-even 6**
2. **Sell 20 cars, break-even 15**
3. **Sell 9 cars, break-even 6**

Why: Sales minus break-even: 8, 5 and 3 cars. The most sales is not the safest.

Task 4

ROUND 2

A deal with a set-up cost.

Profit on the deal?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Livery and set-up, one-off	**£7,000**
Customer pays per car	**£10,750**
You pay per car	**£9,000**
They order	**9 cars**

METHOD

Profit = cars ordered × (customer pays − you pay) − set-up

9 × (£10,750 − £9,000) − £7,000 = £8,750

Why: The first cars pay the set-up; the rest is profit: £8,750.

Task 5

ROUND 3

Both deals are for 10 cars. The taxi deal breaks even at 4 cars, the hire deal at 8.

Which deal is less risky, and why?

PICK A OR B AND GIVE A REASON

- ○ The hire deal
- ✔ **The taxi deal**

Why: The taxi deal's margin of safety is 6 cars; the hire deal's only 2.

Task 6

ROUND 3

Read the chart at 5 cars.

Profit at the cars you expect to sell? Minus allowed.

WORK OUT THE NUMBER · EXAMPLE FIGURES

Fixed costs	**£13,250**
Total costs at 5 cars	**£50,750**
Total revenue at 5 cars	**£50,000**
Cars you expect to sell	**9 cars**

METHOD

Profit = cars × (price – cost per car), read from the chart, – fixed costs

(£50,750 – £13,250) ÷ 5 = £7,500 a car; £50,000 ÷ 5 = £10,000; 9 × £2,500 – £13,250 = £9,250

Why: Each car adds £2,500 above its own cost; the fixed costs come off once.

Final questions

6
questions for each pupil

150
Level 2 Business questions

0
of them on break-even (financial calculations)

No question in the pool is about break-even (financial calculations) yet. The game draws 6 of the 150 Level 2 questions.